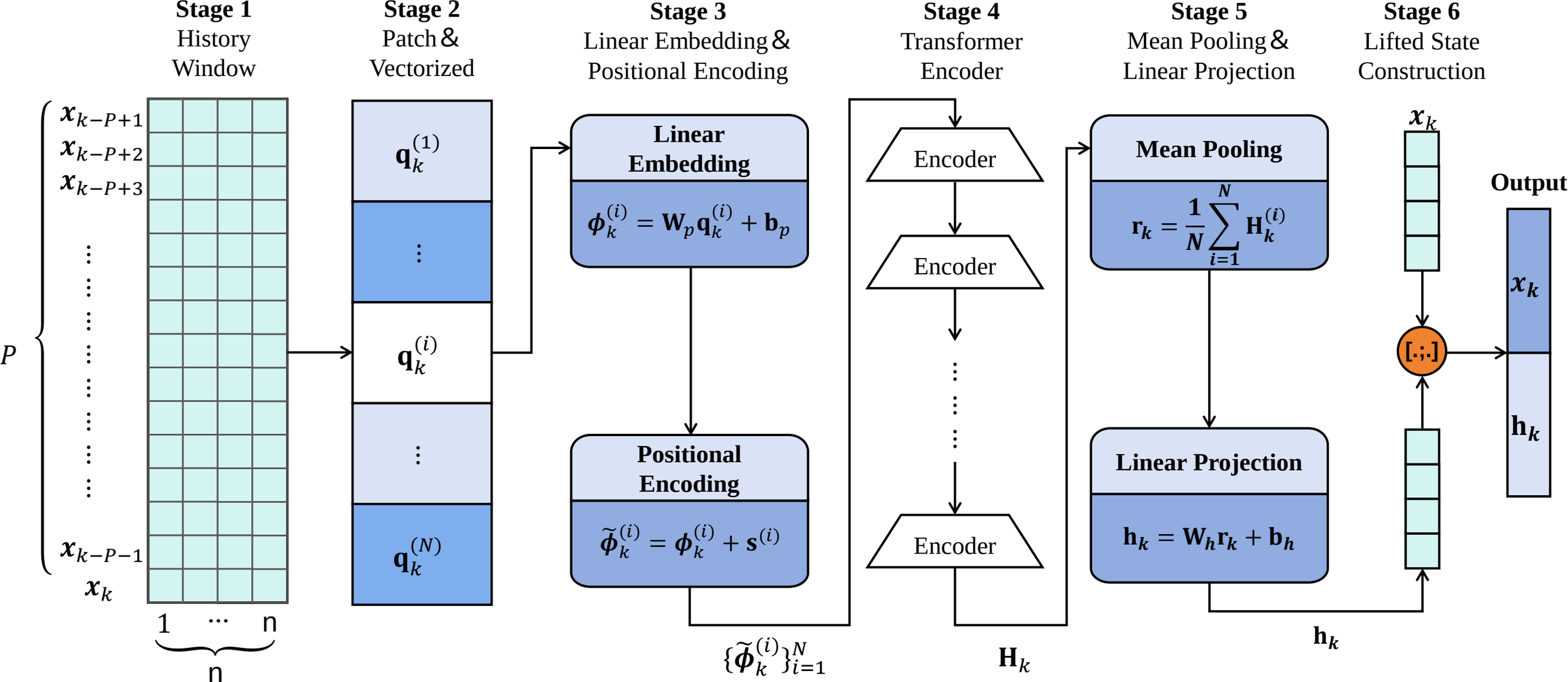

Stage 1
History Window
Stage 2
Patch & Vectorized
Stage 3
Linear Embedding & Positional Encoding
Stage 4
Transformer Encoder
Stage 5
Mean Pooling & Linear Projection
Stage 6
Lifted State Construction
x_{k-P+1}
x_{k-P+2}
x_{k-P+3}
x_{k-P-1}
x_k
P
1 ⋯ n
n
$\mathbf{q}_k^{(1)}$
$\mathbf{q}_k^{(i)}$
$\mathbf{q}_k^{(N)}$
Linear Embedding
$\boldsymbol{\phi}_k^{(i)} = \mathbf{W}_p \mathbf{q}_k^{(i)} + \mathbf{b}_p$
Positional Encoding
$\tilde{\boldsymbol{\phi}}_k^{(i)} = \boldsymbol{\phi}_k^{(i)} + \mathbf{s}^{(i)}$
$\{\tilde{\boldsymbol{\phi}}_k^{(i)}\}_{i=1}^N$
Encoder
Encoder
Encoder
$\mathbf{H}_k$
Mean Pooling
$\mathbf{r}_k = \frac{1}{N}\sum_{i=1}^{N} \mathbf{H}_k^{(i)}$
Linear Projection
$\mathbf{h}_k = \mathbf{W}_h \mathbf{r}_k + \mathbf{b}_h$
$\mathbf{h}_k$
x_k
[.;.]
Output
x_k
$\mathbf{h}_k$